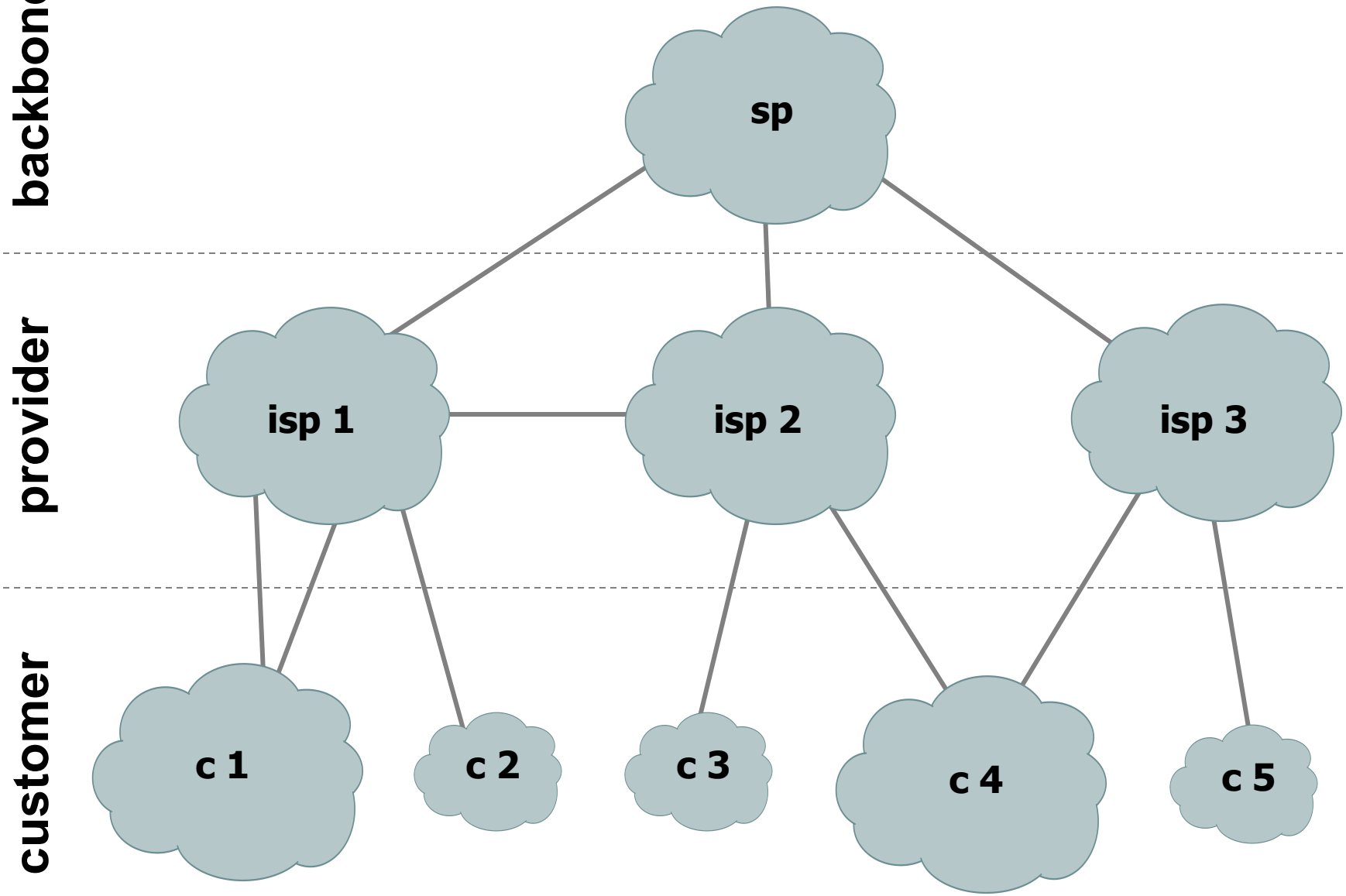
Advanced Computer networks

2025-2026

BGP - stub networks

Michele Bocca Garetto - based on slides by: kathara.org

a small internet



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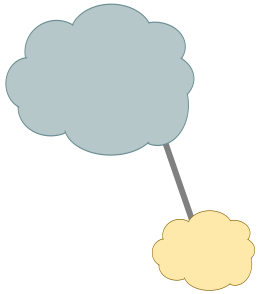
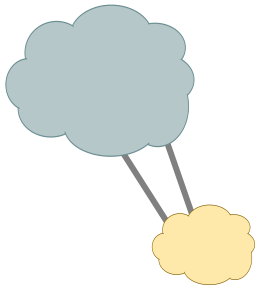
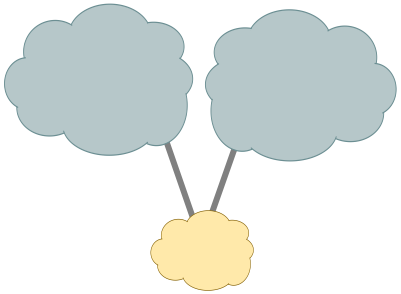
customer classification



- stub networks
  - one link to a single isp



- multi-homed stub network
  - two or more links to the same isp
  - purposes: backup or load sharing



- multi-homed network
  - two or more links to different isps
  - purposes: backup or load sharing

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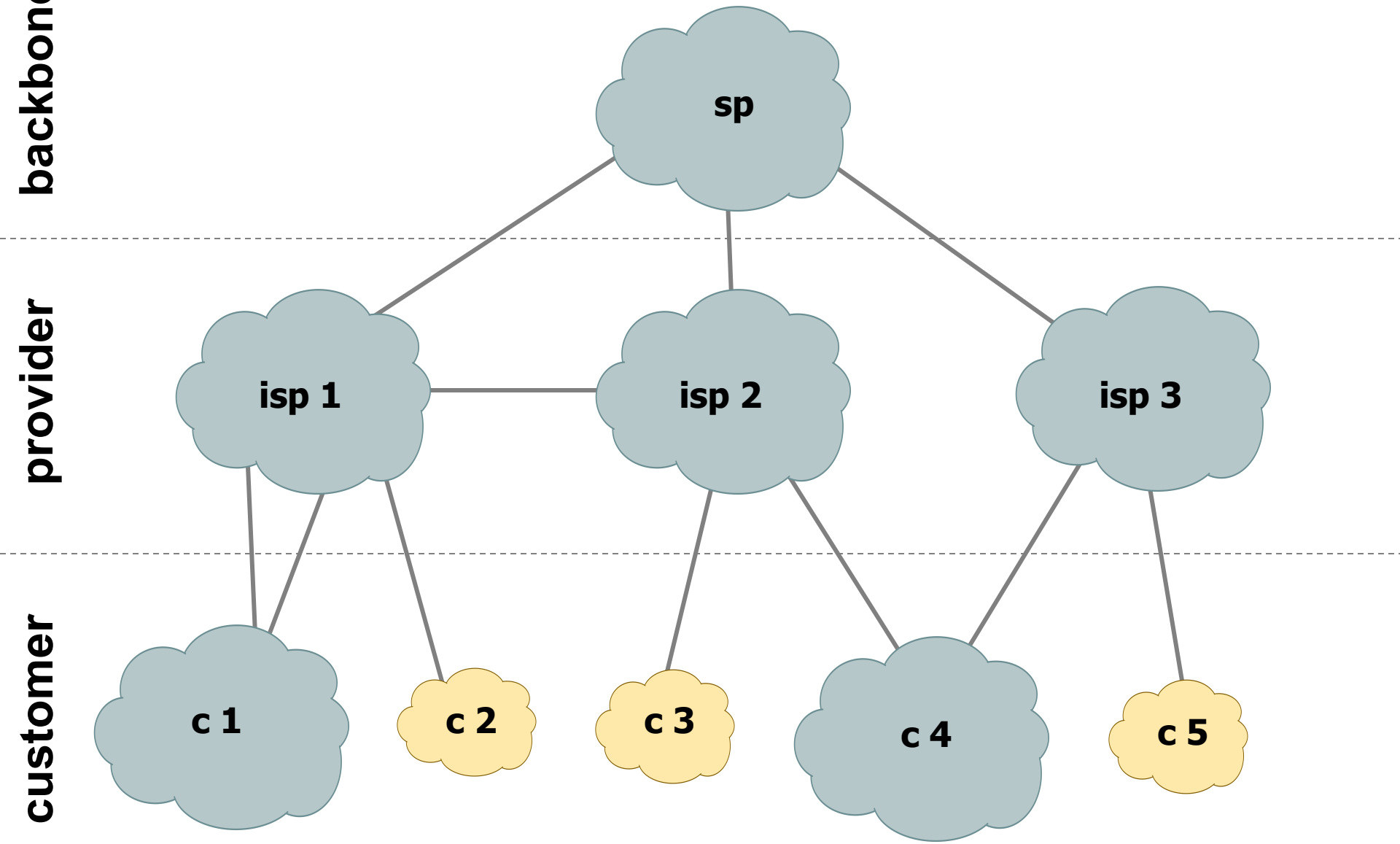
Stub networks

stub networks



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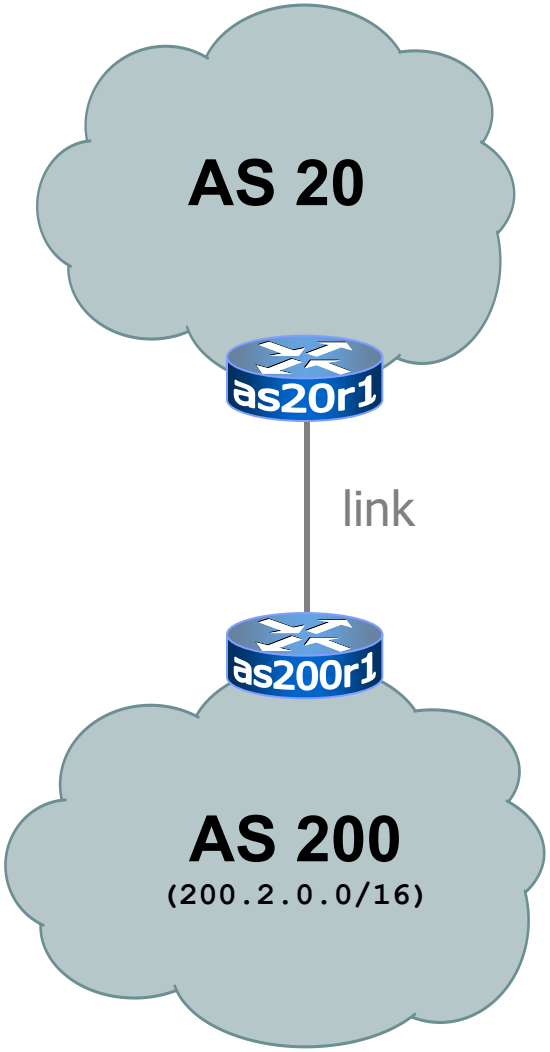
stub network architecture



- one of the customer routers is chosen to be the default gateway
- the router is attached to a single router of the isp with a link (possibly more than one)
- a single peering in which as200 announces its route and accepts the default is enough

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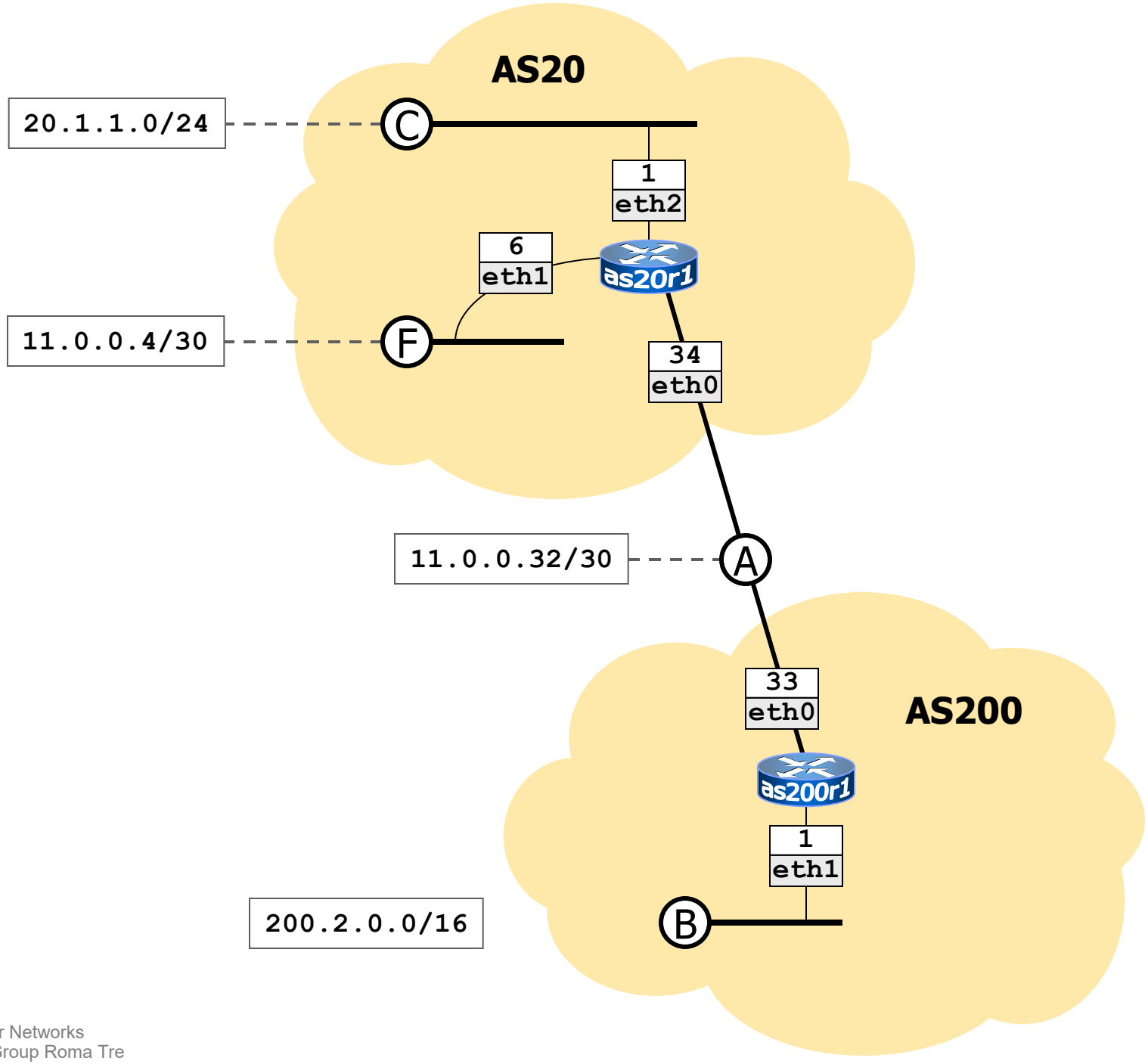

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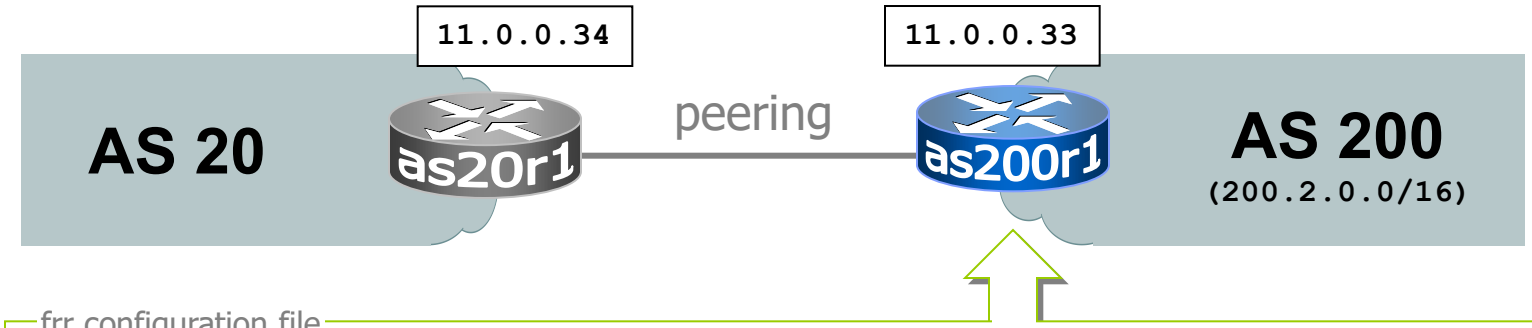
router as200r1 configuration



frr configuration file

```
! router as200r1 (customer side)
!
router bgp 200
no bgp ebgp-requires-policy
network 200.2.0.0/16
neighbor 11.0.0.34 remote-as 20
neighbor 11.0.0.34 description Router as20r1
```

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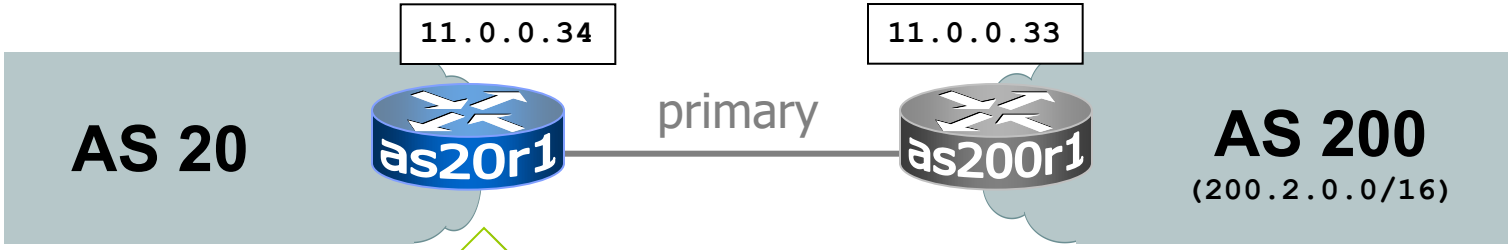
router as20r1 configuration



frr configuration file

```
! router as20r1 (isp side)
router bgp 20
no bgp network import-check
neighbor 11.0.0.33 remote-as 200
neighbor 11.0.0.33 description Router as200r1
neighbor 11.0.0.33 default-originate
neighbor 11.0.0.33 prefix-list customerIn in
neighbor 11.0.0.33 prefix-list defaultOut out
!
network 20.1.1.0/24
network 0.0.0.0/0
!
ip prefix-list customerIn permit 200.2.0.0/16
ip prefix-list defaultOut permit 0.0.0.0/0
```

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about `network 0.0.0.0/0`

- using `network 0.0.0.0/0` is enough to
  - place a default route in the local bgp routing table
    - as with any other prefix, the network command does not injected the default into the kernel
  - announce it
    - provided that you have it in the kernel or that you have a "`no bgp network import-check`" configuration
    - provided that you have set filters on the outgoing announces to the neighbors or that you have a "`no bgp ebgp-requires-policy`" configuration

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about `network 0.0.0.0/0`

- it makes sense using `network 0.0.0.0/0` at the top of the isp hierarchy to originate the default route and to propagate it to all downstream providers
  - the default will be propagated with the full AS-path
- `network 0.0.0.0/0` should not be used at intermediate levels of the hierarchy
  - otherwise, routers would prefer the locally originated default route and remove the one offered by their upstream from the forwarding table

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about `default originate`

- using **`default-originate`** for a specific neighbor
  - does not place a default route in the local bgp routing table
    - nor in the local kernel routing table
  - announces the default route to that neighbor
    - the prefix seems to be originated from the announcing AS as if it was declared with **`network`** `0.0.0.0/0` even if it is not
    - **`default-originate`** overwrites the possible present **`network`** `0.0.0.0/0` declaration

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default-originate and route-maps

- a default route originated with **network** 0.0.0.0/0 is handled like any other route
  - **route-maps** used with a specific neighbor are applied to the default route as well
- a default route originated with **default-originate** is processed by a different **route-map**:

command syntax

```
neighbor <neighbor-ip> default-originate route-map
   <r-map-name> in
```

command syntax

```
neighbor <neighbor-ip> default-originate route-map
   <r-map-name> out
```

stub as: lab

- start the lab

host machine

```
user@localhost:~$ cd kathara-lab_bgp-stub-as_frr
user@localhost:~/kathara-lab_bgp-stub-as_frr$ kathara lstart
```

- check the frr configuration file

as20r1

```
as20r1:~# less /etc/frr/frr.conf
```

- check the frr log file

as20r1

```
as20r1:~# less /var/log/frr/frr.log
```

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stub as: lab

- check the routing table of as20r1

as20r1

```
root@as20r1:/# vtysh

Hello, this is FRRouting (version 7.5.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

as20r1-frr# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

C>* 11.0.0.4/30 is directly connected, eth1, 00:14:46
C>* 11.0.0.32/30 is directly connected, eth0, 00:14:46
C>* 20.1.1.0/24 is directly connected, eth2, 00:14:46
B>* 200.2.0.0/16 [20/0] via 11.0.0.33, eth0, weight 1, 00:14:43
as20r1-frr#
```

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stub as: lab

- check the bgp status

as20r1

```
as20r1-frr# show ip bgp neighbors
BGP neighbor is 11.0.0.33, remote AS 200, local AS 20, external link
 Description: Router as200r1
Hostname: as200r1
.....
as20r1-frr# show ip bgp
BGP table version is 3, local router ID is 20.1.1.1, vrf id 0
Default local pref 100, local AS 20
Status codes:  s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes:  i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 0.0.0.0/0        0.0.0.0                  0         32768 i
*> 20.1.1.0/24      0.0.0.0                  0         32768 i
*> 200.2.0.0/16     11.0.0.33                0             0 200 i

Displayed  3 routes and 3 total paths
as20r1-frr#
```

stub as: lab

- perform several pings on the routers
- terminate the lab

host machine

```
user@localhost:~/kathara-lab_bgp-stub-as_frr$ kathara lclean
```

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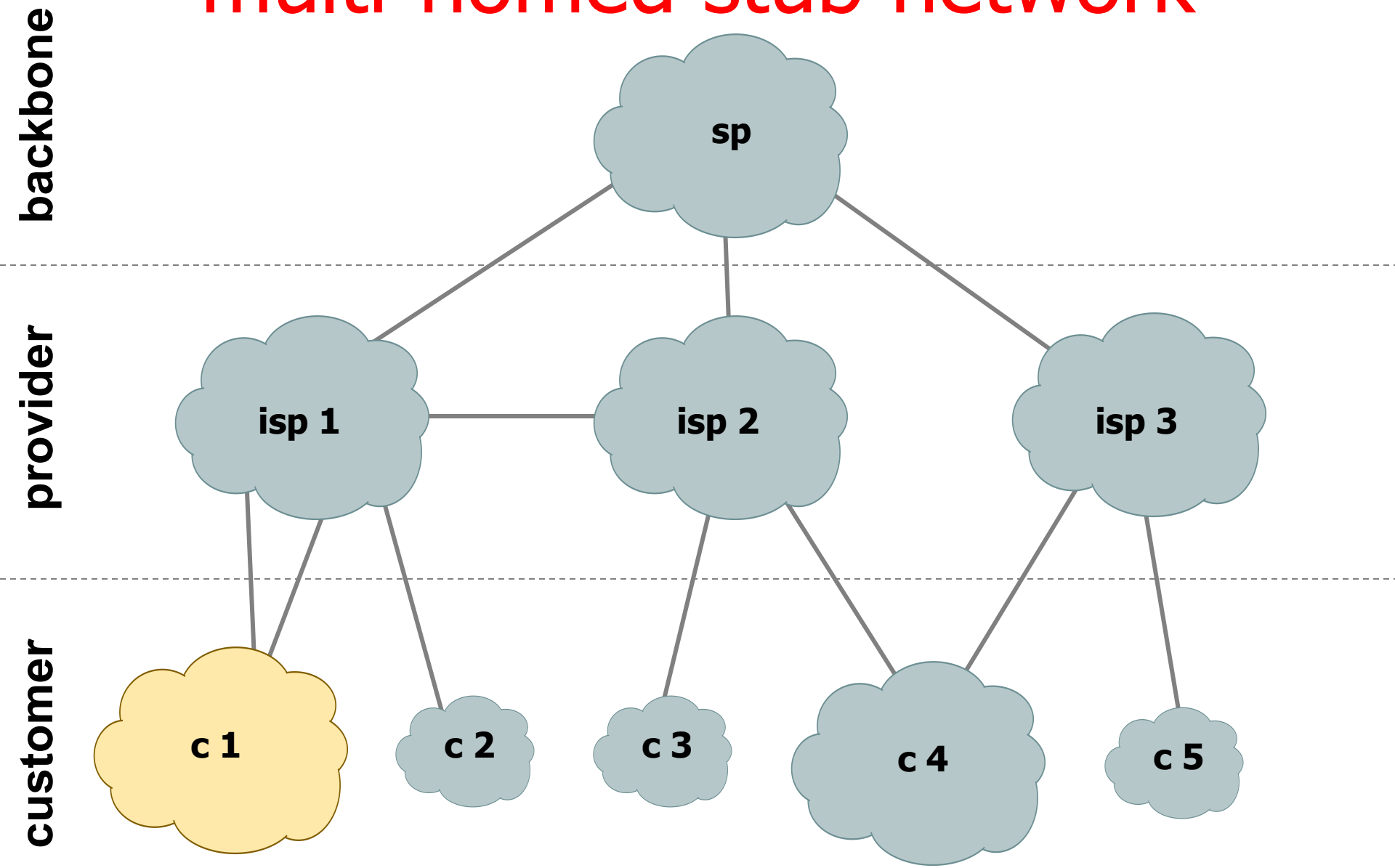
Multi-homed Stub networks

multi-homed stub network



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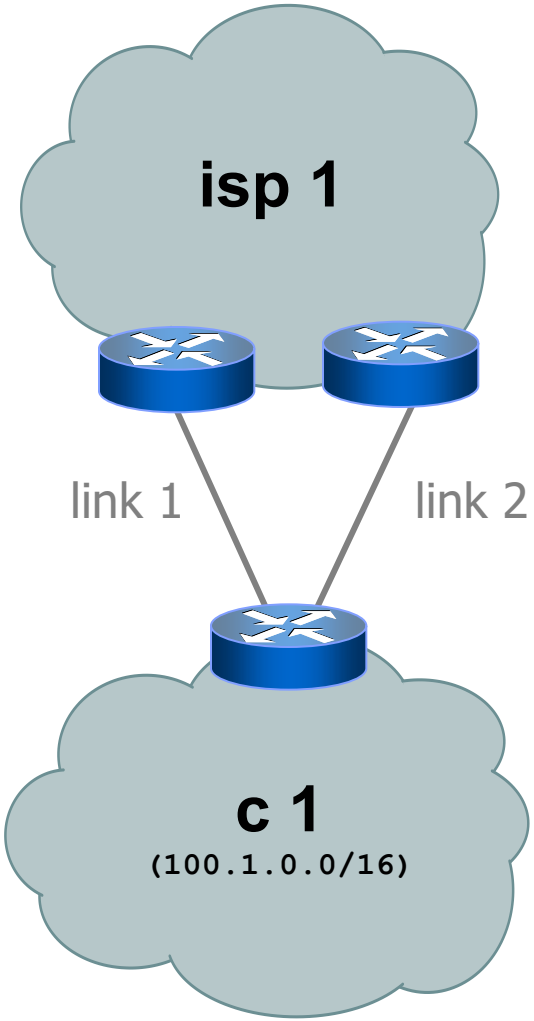
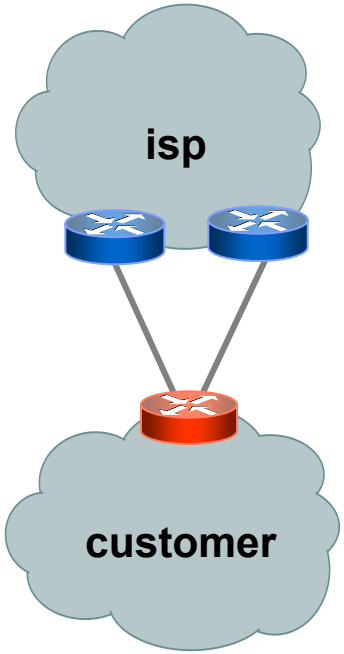
multi-homed stub network

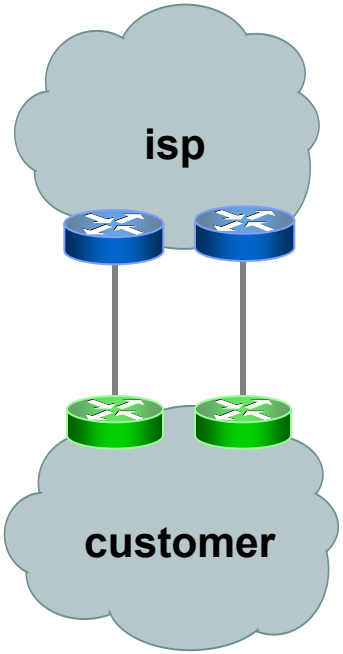


- two links to the same isp
- generally two routers of the customer as are involved

single point of failure



augmented redundancy



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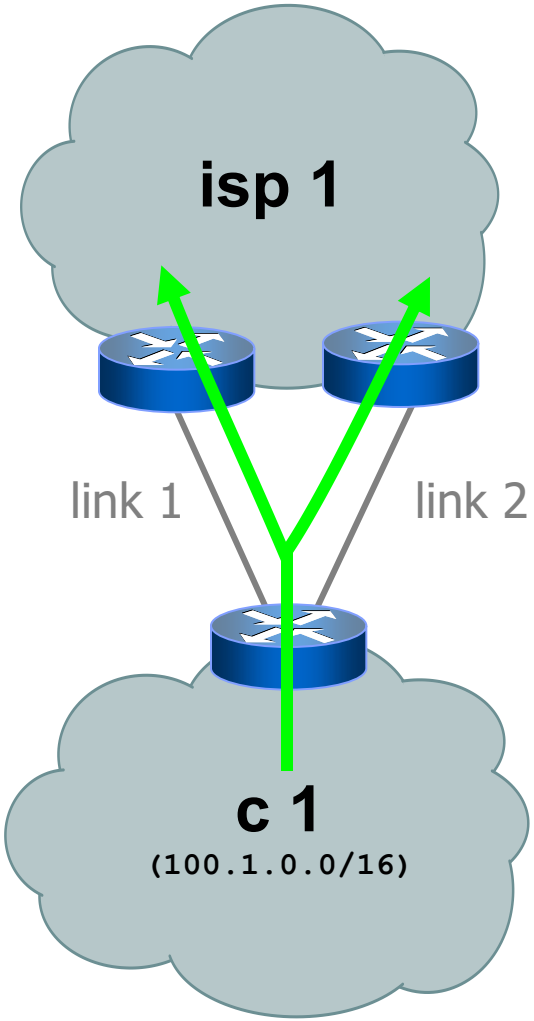
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet

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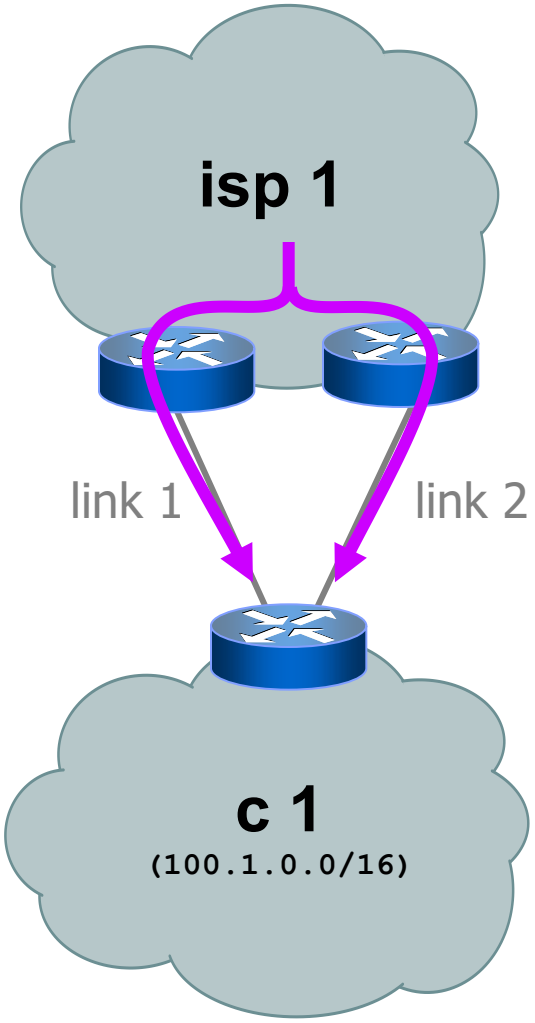
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet
- an inbound packet may use any of the two links in order to reach the network

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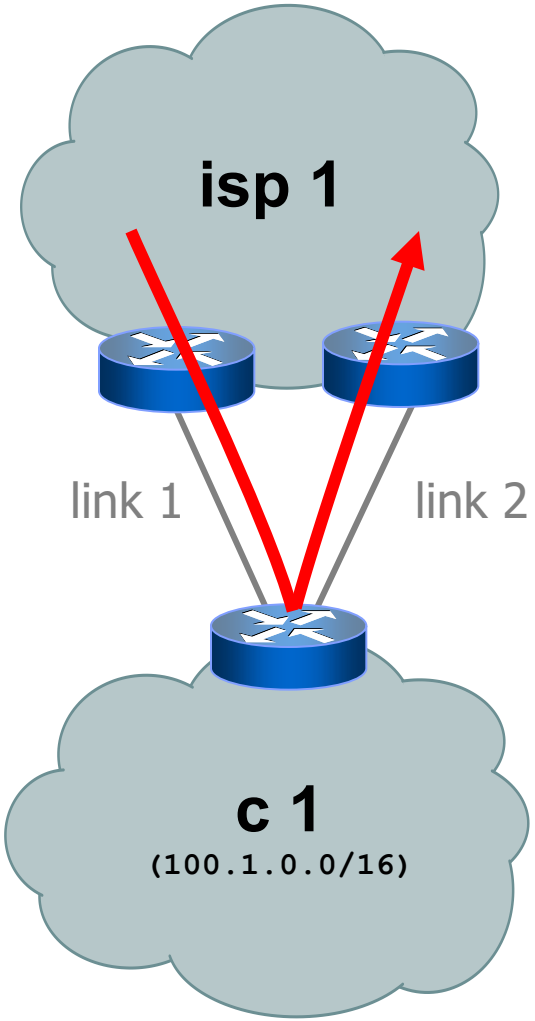
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet
- an inbound packet may use any of the two links in order to reach the network
- an internet packet may traverse link 1 and link 2 (or vice versa)

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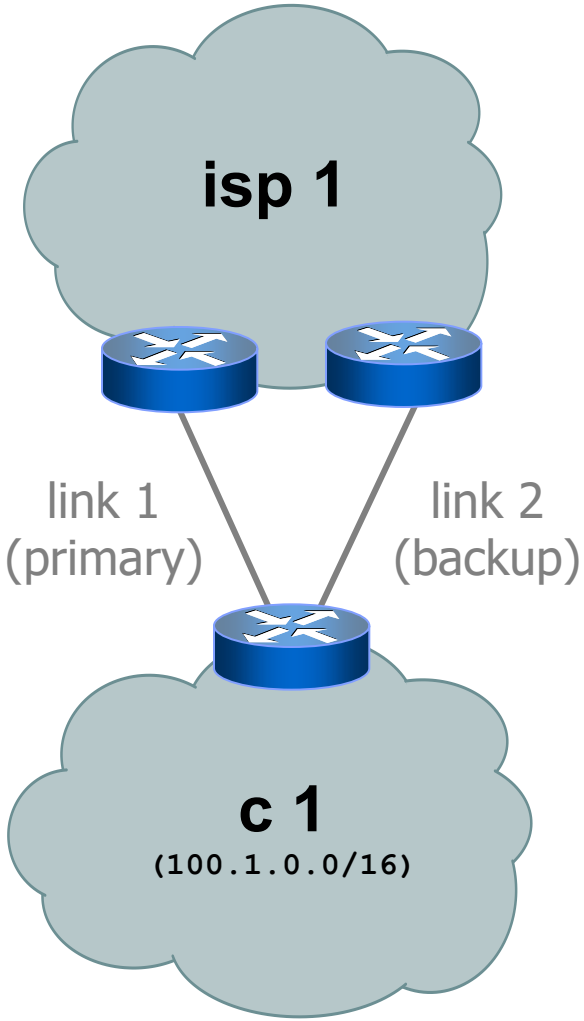
desired policy: backup



- rule out transit flows
- inbound traffic:
  - use link 1
  - use link 2 when link 1 is unavailable
- outbound traffic:
  - use link 1
  - use link 2 when link 1 is unavailable

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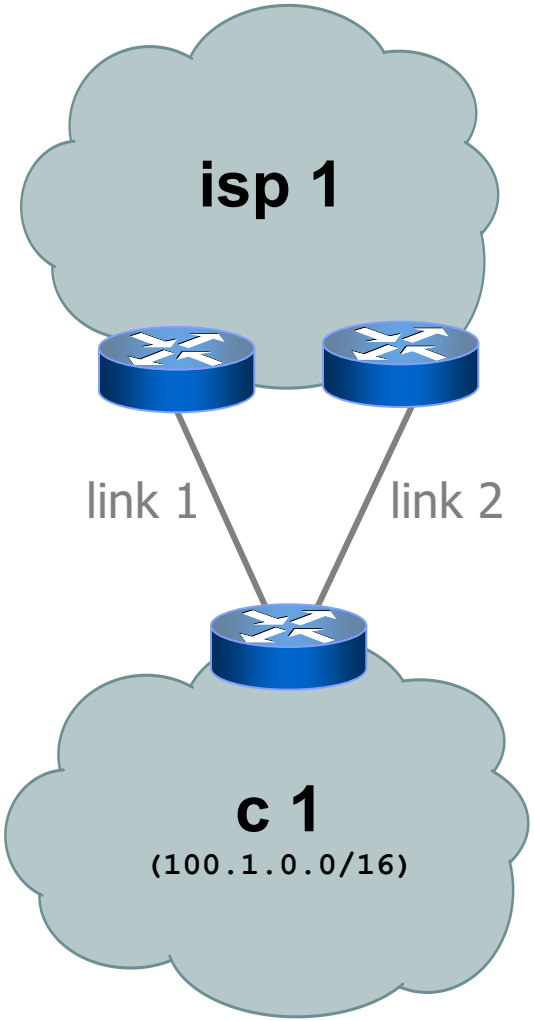
alternatives to using bgp



- using an igp (is-is, ospf, rip,...)
  - packets use link 1 or link 2 depending on the shortest path to customer c 1
  - there is no way to rule out transit packets when link 1 and link 2 are on the minimum path between a source and a destination
- using static routes
  - both the routers of the isp and the network have to be coherently configured by hand
  - there is no way to manage an automatic backup mechanism

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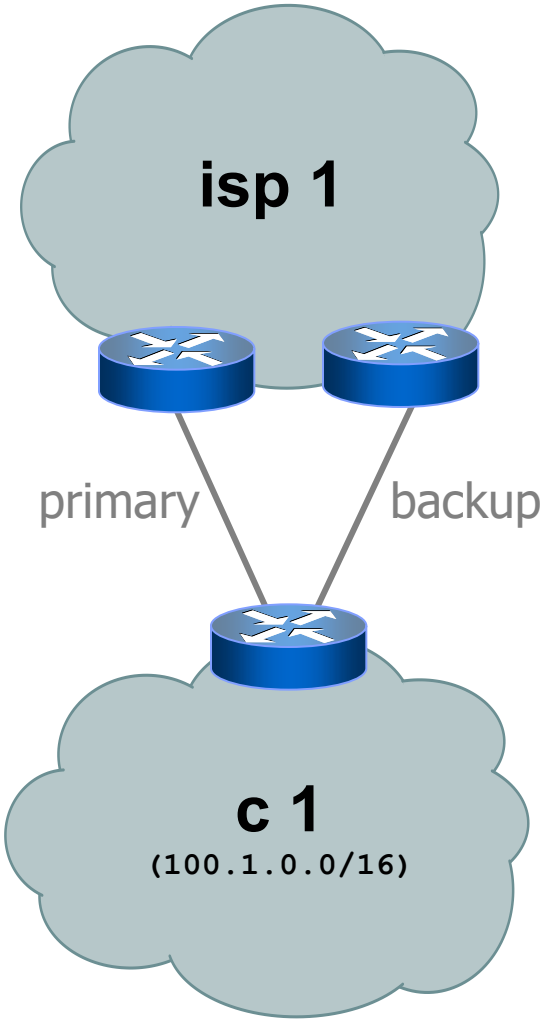
using bgp



- announce /16 aggregate on each link
  - primary link makes standard announcements
  - backup link increases metric on outbound announcements, and reduces local-pref on inbound announcements
- when one link fails, the announcement of the /16 aggregate via the other link ensures continued connectivity

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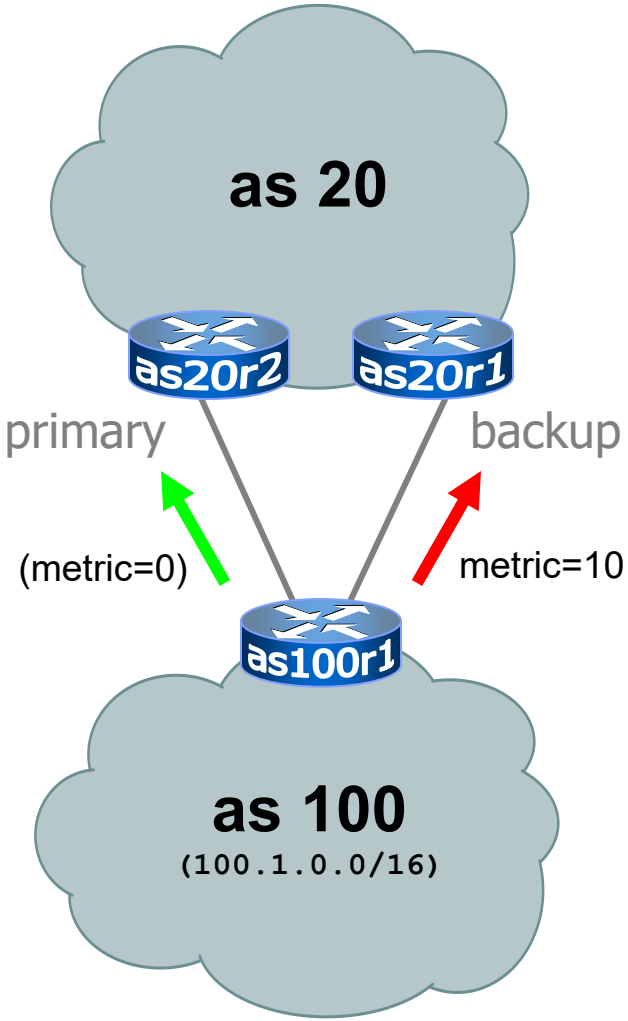
setting metric



- the value of the "multi-exit-discriminator" attribute is called "metric"
- upon receiving the same announcement with two different meds, the provider will (hopefully) adopt the one with the smaller one
- the metric is set on outgoing announcements and manages inbound traffic flows
- metrics are comparable only among announcements coming from the same neighboring as

default value: 0

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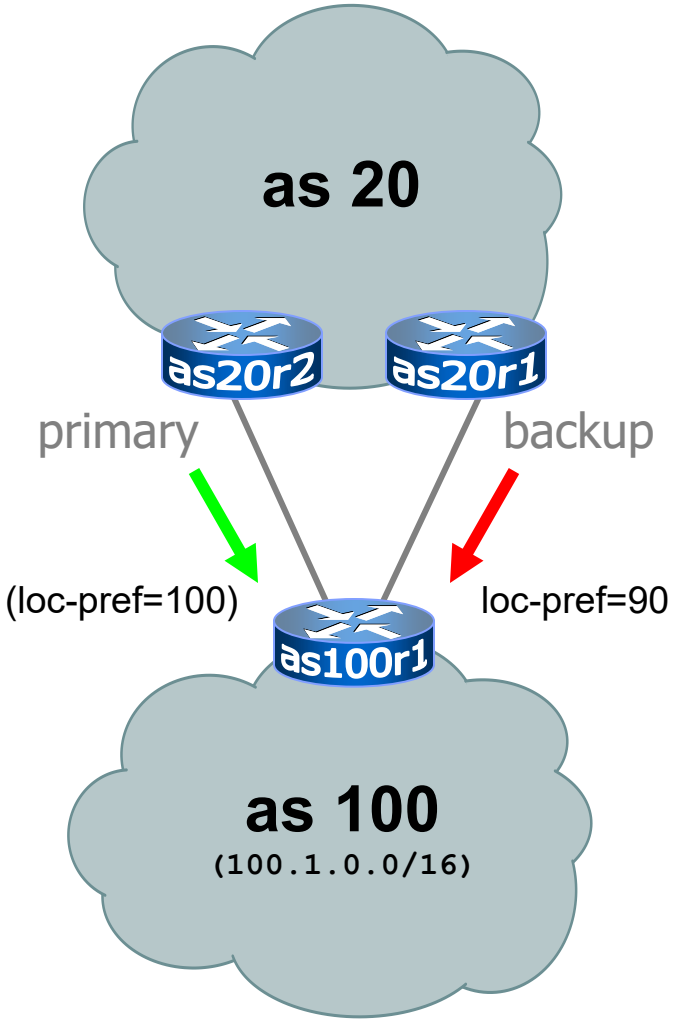
setting local-preference



- the customer assigns a lower local-preference to the announcement coming from the backup peer
- the local-preference attribute is checked before as-path length in the route selection process
- local-preference applies to incoming announcements and manages outbound traffic flows

default value: 100

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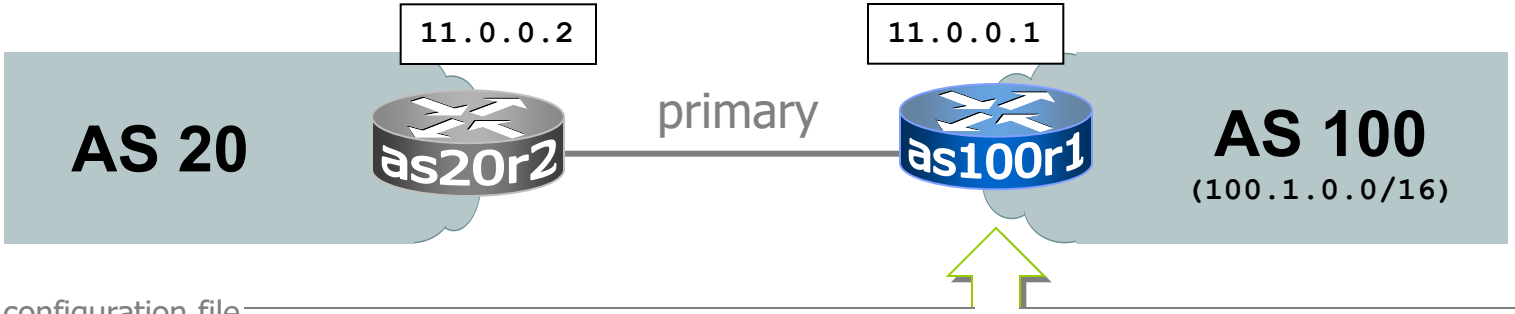
router as100r1 configuration



frr configuration file

```
router bgp 100
!
neighbor 11.0.0.2 remote-as 20
neighbor 11.0.0.2 description as20r2 (primary)
neighbor 11.0.0.6 remote-as 20
neighbor 11.0.0.6 description as20r1 (backup)
!
network 100.1.0.0/16
!
… next slide
```

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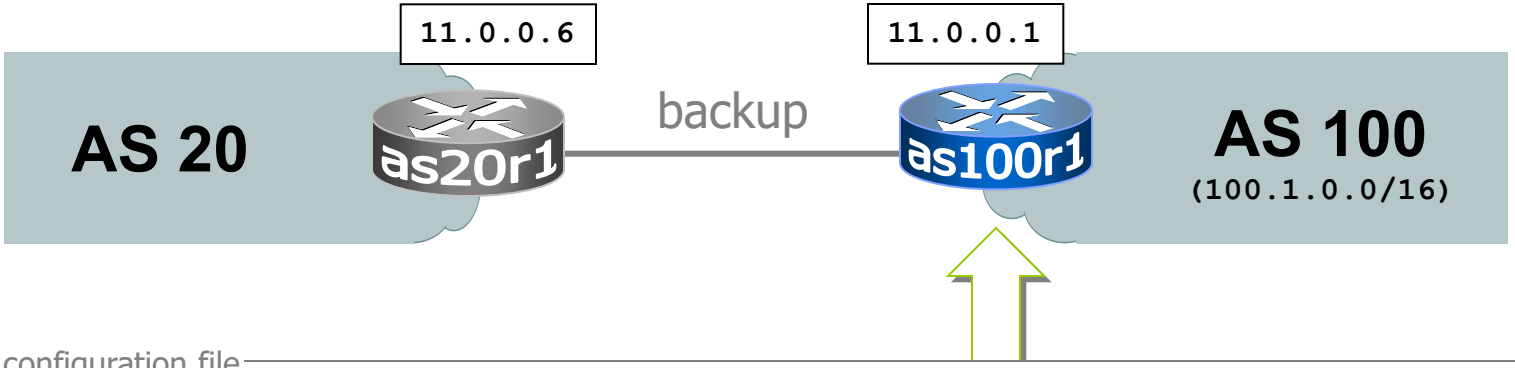
router as100r1 configuration



frr configuration file

```
!
neighbor 11.0.0.2 prefix-list mineOutOnly out
neighbor 11.0.0.2 prefix-list defaultIn in
neighbor 11.0.0.6 prefix-list mineOutOnly out
neighbor 11.0.0.6 route-map metricOut out
neighbor 11.0.0.6 prefix-list defaultIn in
neighbor 11.0.0.6 route-map localPrefIn in
!
… next slide
```

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router as100r1 configuration

frr configuration file

```
access-list myAggregate permit 100.1.0.0/16
!
ip prefix-list mineOutOnly permit 100.1.0.0/16
ip prefix-list defaultIn permit 0.0.0.0/0
!
route-map metricOut permit 10
match ip address myAggregate
set metric 10
!
route-map localPrefIn permit 10
set local-preference 90
!
```

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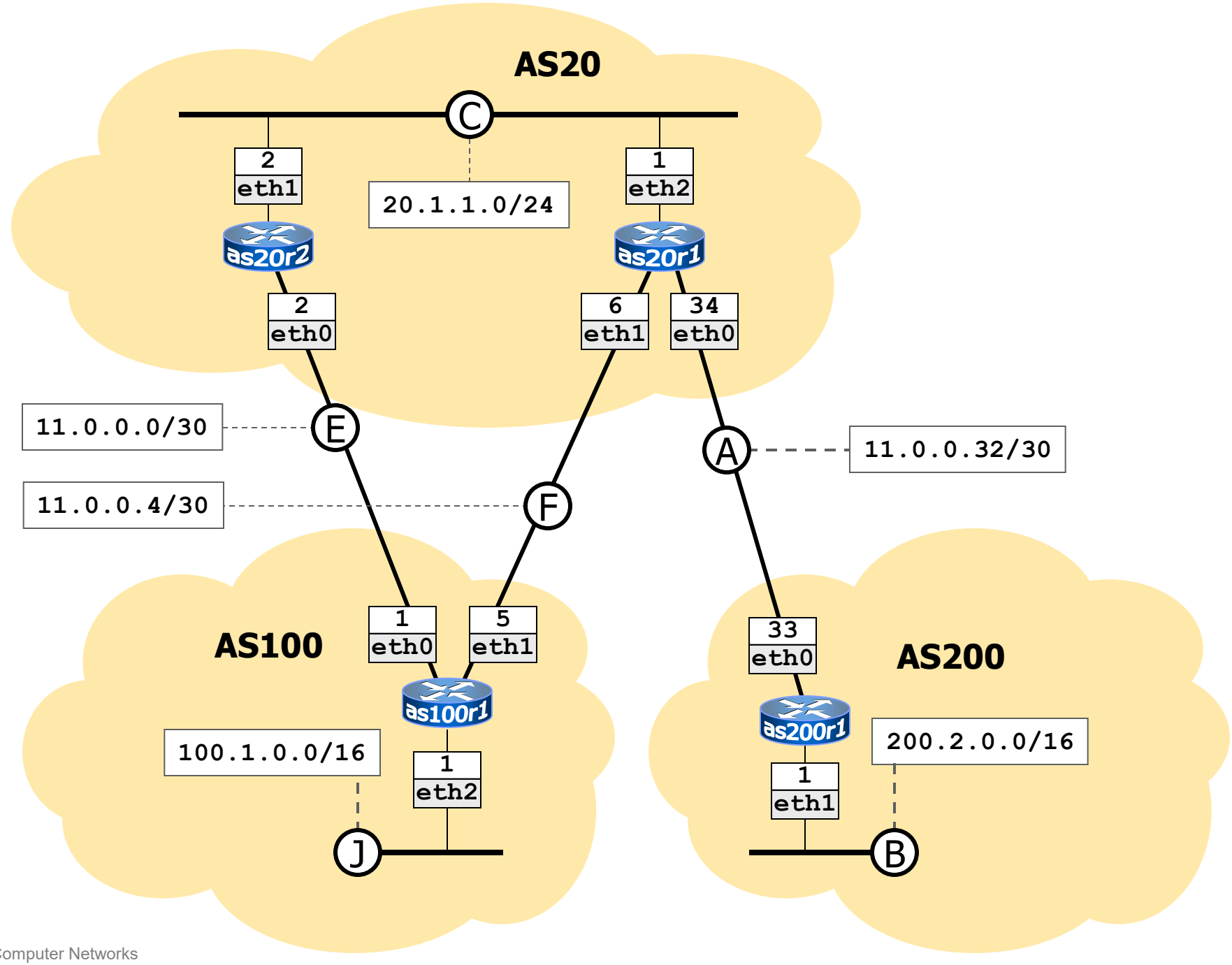

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multi-homed stub

- start the lab

host machine

```
user@localhost:~$ cd kathara-lab_bgp-multi-homed-stub_frr
user@localhost:~/kathara-lab_bgp-multi-homed-stub_frr$ kathara lstart
```

- ping as100r1 from as200r1

as200r1

```
root@as200r1:~# ping 100.1.0.1
PING 100.1.0.1 (100.1.0.1) 56(84) bytes of data.
64 bytes from 100.1.0.1: icmp_seq=1 ttl=62 time=1.39 ms
64 bytes from 100.1.0.1: icmp_seq=2 ttl=62 time=1.88 ms

--- 100.1.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1022ms
rtt min/avg/max/mdev = 1.398/1.642/1.886/0.244 ms
```

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multi-homed stub

- traceroute from as200r1 to as100r1

as200r1

```
root@as200r1:/# traceroute 100.1.0.1
traceroute to 100.1.0.1 (100.1.0.1), 30 hops max, 60 byte packets
 1  11.0.0.34 (11.0.0.34)  6.665 ms  6.222 ms  0.129 ms
 2  20.1.1.2 (20.1.1.2)  0.214 ms  0.179 ms  0.170 ms
 3  100.1.0.1 (100.1.0.1)  0.277 ms  0.208 ms  0.210 ms
root@as200r1:/#
```

- the packets traverse router as20r2 (20.1.1.2) to reach 100.1.0.1
  - the backup link is not used

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multi-homed stub

- let us have a look at frr bgp table

as20r1

```
root@as20r1:~# vtysh
.....
as20r1-frr> show ip bgp
BGP table version is 6, local router ID is 20.1.1.1, vrf id 0
Default local pref 100, local AS 20
Status codes:  s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes:  i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
   0.0.0.0/0        0.0.0.0                  0         32768 i
*>i11.0.0.0/30      20.1.1.2                 0    100      0 i
*> 11.0.0.4/30      0.0.0.0                  0         32768 i
*> 11.0.0.32/30     0.0.0.0                  0         32768 i
* i20.1.1.0/24      20.1.1.2                 0    100      0 i
*>                  0.0.0.0                  0         32768 i
*  100.1.0.0/16     11.0.0.5                10             0 100 i
*>i                 11.0.0.1                 0    100      0 100 i
*> 200.2.0.0/16     11.0.0.33                0             0 200 i
Displayed  7 routes and 9 total paths
```

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recursive lookup problem

- some implementations of bgp would not accept a route if the next-hop is not known via igp
- this below is an example from the analogous lab using quagga

as20r1

```
as20r1:~# telnet localhost bgpd
.....
bgpd> show ip bgp
BGP table version is 0, local router ID is 20.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
.....
*>i11.0.0.0/30      20.1.1.2                 0    100      0 i
.....
* i100.1.0.0/16     11.0.0.1                 0    100      0 100 i
*>                  11.0.0.5                10             0 100 i
.....
Total number of prefixes 7
```

not accepted by quagga because next hop is only known via bgp and not via igp

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multi-homed stub

- now shut down the primary connection on as100r1

as100r1

```
root@as100r1:~# vtysh

Hello, this is FRRouting (version 7.5.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

as100r1-frr# configure terminal
as100r1-frr(config)# router bgp 100
as100r1-frr(config-router)# neighbor 11.0.0.2 shutdown
as100r1-frr(config-router)# quit
as100r1-frr(config)# quit
as100r1-frr# show ip bgp summary

IPv4 Unicast Summary:
BGP router identifier 100.1.0.1, local AS number 100 vrf-id 0
BGP table version 3
RIB entries 2, using 384 bytes of memory
Peers 2, using 43 KiB of memory

Neighbor   V   AS   MsgRcvd   MsgSent TblVer  InQ OutQ   Up/Down State/PfxRcd   PfxSnt
11.0.0.2   4   20        43        45      0    0    0 00:00:20 Idle (Admin)        0
11.0.0.6   4   20        44        44      0    0    0 00:39:33            1        1

Total number of neighbors 2
as100r1-frr#
```

multi-homed stub

- check the backup

as100r1

```
root@as100r1:/# route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         11.0.0.6        0.0.0.0         UG    20     0        0 eth1
11.0.0.0        0.0.0.0         255.255.255.252 U     0      0        0 eth0
11.0.0.4        0.0.0.0         255.255.255.252 U     0      0        0 eth1
100.1.0.0       0.0.0.0         255.255.0.0     U     0      0        0 eth2
```

as100r1

```
root@as100r1:/# traceroute 200.2.0.1
traceroute to 200.2.0.1 (200.2.0.1), 30 hops max, 60 byte packets
 1  11.0.0.6 (11.0.0.6)  5.575 ms  5.158 ms  4.913 ms
 2  200.2.0.1 (200.2.0.1)  4.720 ms  4.448 ms  4.207 ms
```

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multi-homed stub

- restart the primary connection and check that the primary link is back

as100r1

```
root@as100r1:/# vtysh

Hello, this is FRRouting (version 7.5.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

as100r1-frr# configure terminal
as100r1-frr(config)# router bgp 100
as100r1-frr(config-router)# no neighbor 11.0.0.2 shutdown
as100r1-frr(config-router)# quit
as100r1-frr(config)# quit
as100r1-frr# quit
root@as100r1:/# traceroute 200.2.0.1
traceroute to 200.2.0.1 (200.2.0.1), 30 hops max, 60 byte packets
 1  11.0.0.2 (11.0.0.2)  4.466 ms  4.230 ms  4.112 ms
 2  20.1.1.1 (20.1.1.1)  4.014 ms  3.867 ms  3.753 ms
 3  200.2.0.1 (200.2.0.1)  3.641 ms  0.234 ms  0.120 ms
root@as100r1:/#
```

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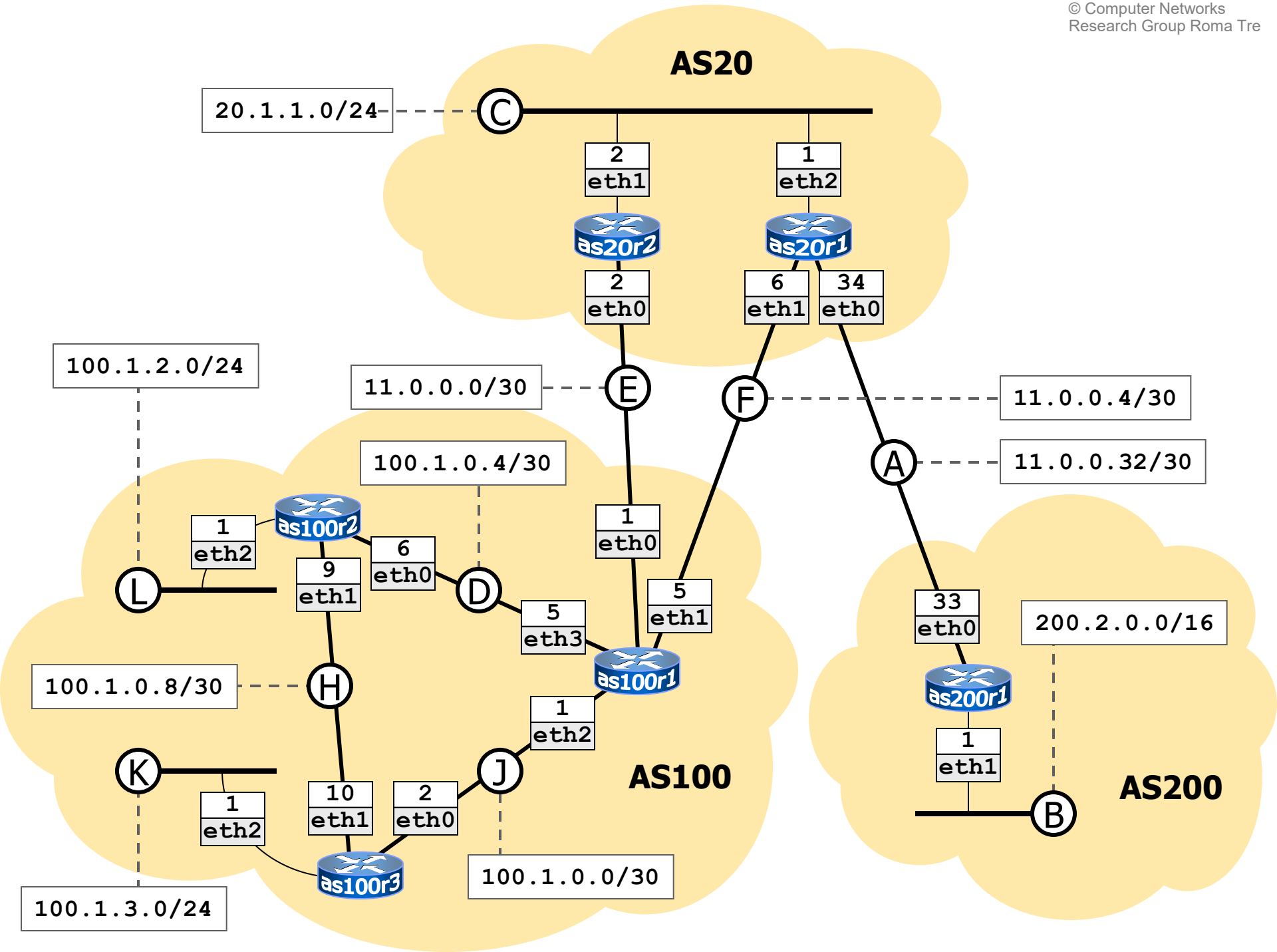

interior gateway protocols

- rip is used:
  - within as20 to propagate reachability information about the next-hops
  - within as100 as an internal routing protocol

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router as100r1 configuration

frr rip configuration file

```
router rip
network 100.1.0.0/16
redistribute bgp
```

- talk rip on some interface
- send distance vector packets through interfaces falling into the specified prefix
- redistribute bgp-learned routes to rip neighbors

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router as100r2 configuration

frr rip configuration file

```
router rip
network 100.1.0.0/16
redistribute connected
```

- talk rip on some interface
- send distance vector packets through interfaces falling into the specified prefix
- redistribute connected networks to rip neighbors
  - the network that is directly connected to a rip enabled interface is automatically inserted in the local rip routing table

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routing table

- rip routing table on as100r1

as100r1

```
as100r1:~# telnet localhost ripd
.....
User Access Verification

Password:
ripd> show ip rip
Codes: R - RIP, C - connected, O - OSPF, B - BGP
       (n) - normal, (s) - static, (d) - default, (r)    edistribute,
       (i) - interface

     Network            Next Hop         Metric From            Time
B(r) 0.0.0.0/0          11.0.0.2              1 self
C(i) 100.1.0.0/30       0.0.0.0               1 self
C(i) 100.1.0.4/30       0.0.0.0               1 self
R(n) 100.1.0.8/30       100.1.0.6             2 100.1.0.6      02:24
R(n) 100.1.2.0/24       100.1.0.6             2 100.1.0.6      02:24
R(n) 100.1.3.0/24       100.1.0.2             2 100.1.0.2      02:41
ripd>
```

route distributed from bgp into rip

play with the network

- have a look at the routing tables of routers inside as100
  - does the 0/0 arrive on as100r2 and as100r3? why?
- try to **`ping`**/**`traceroute`** all the pairs of routers
- look for bgp updates in bgpd logs
- capture (bgp) packets on the routers
- create faults on collision domain E

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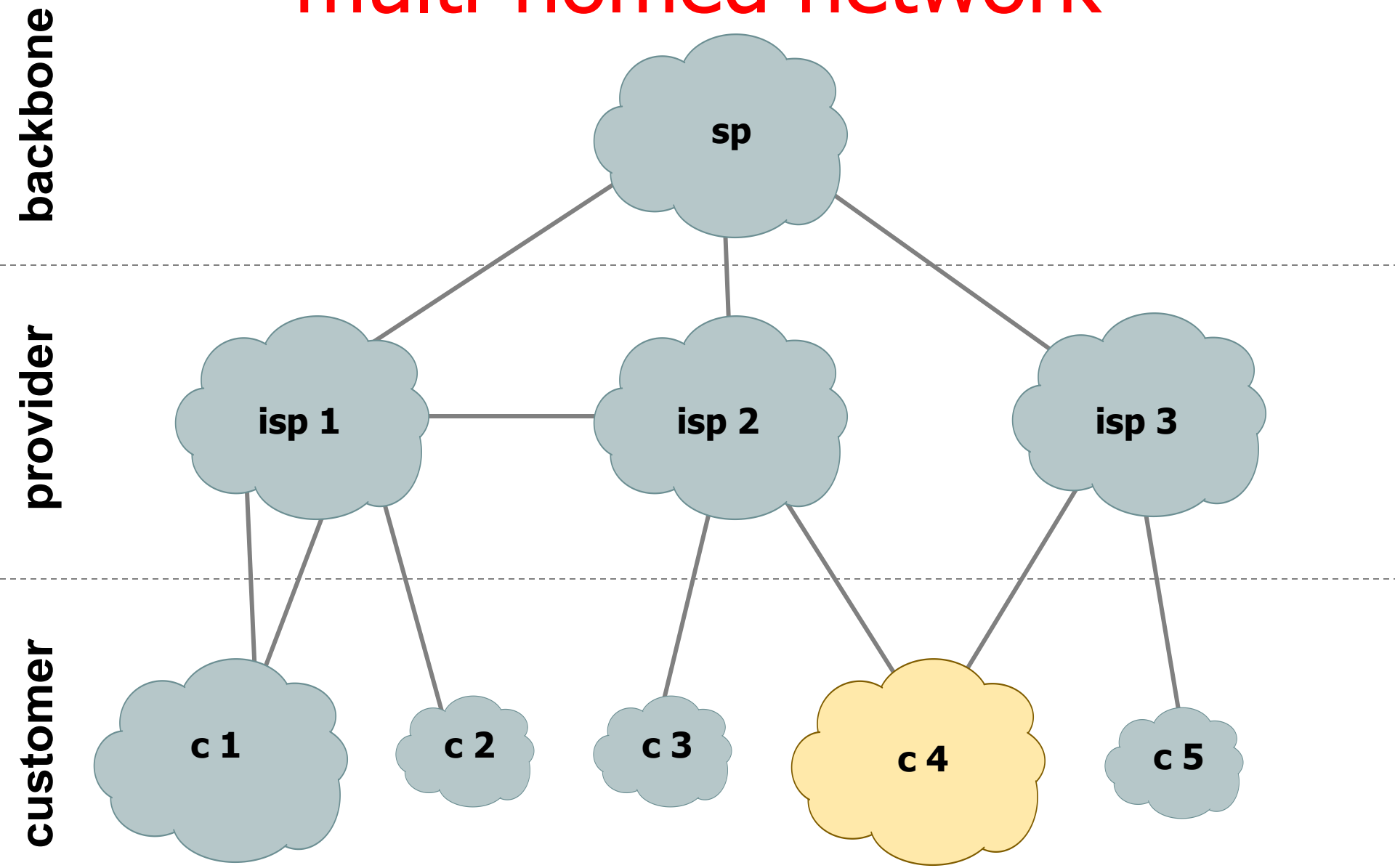
Multi-homed networks

multi-homed network



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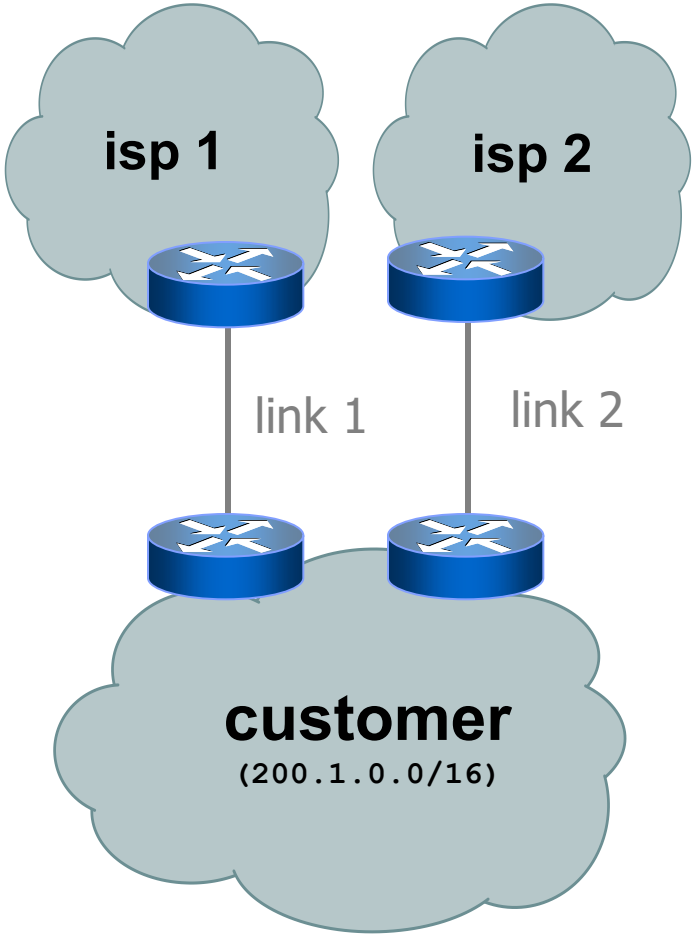
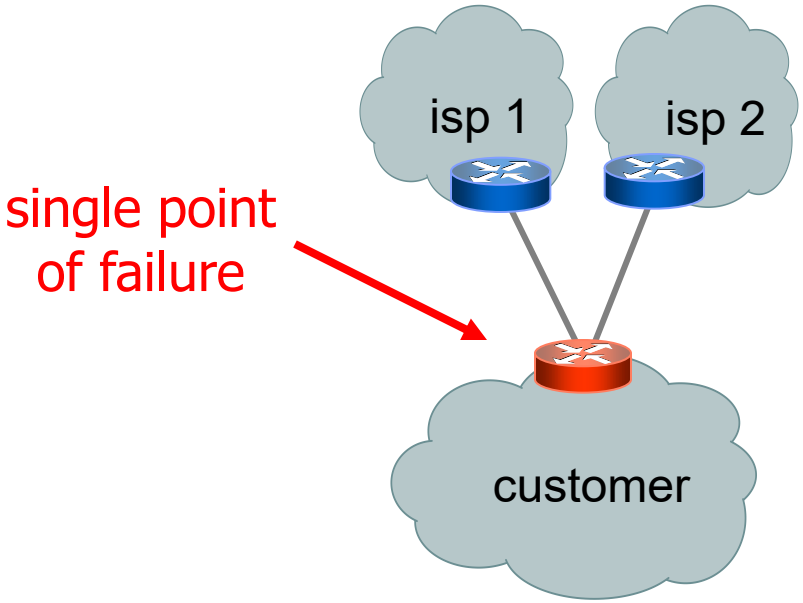
multi-homed network



- two links to two different providers
- generally two routers are involved in order to avoid single points of failures



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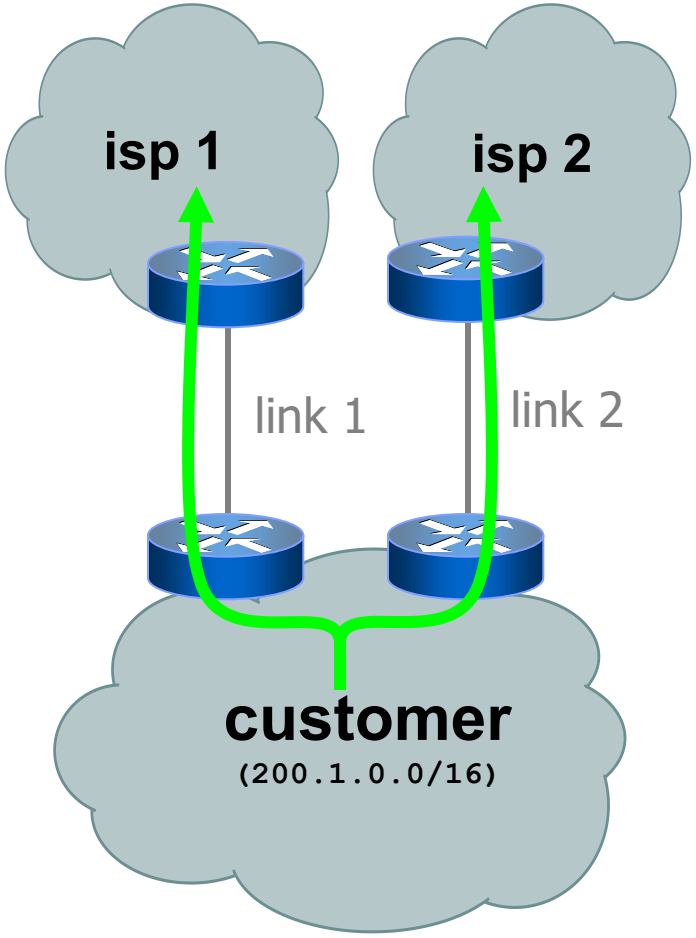
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet

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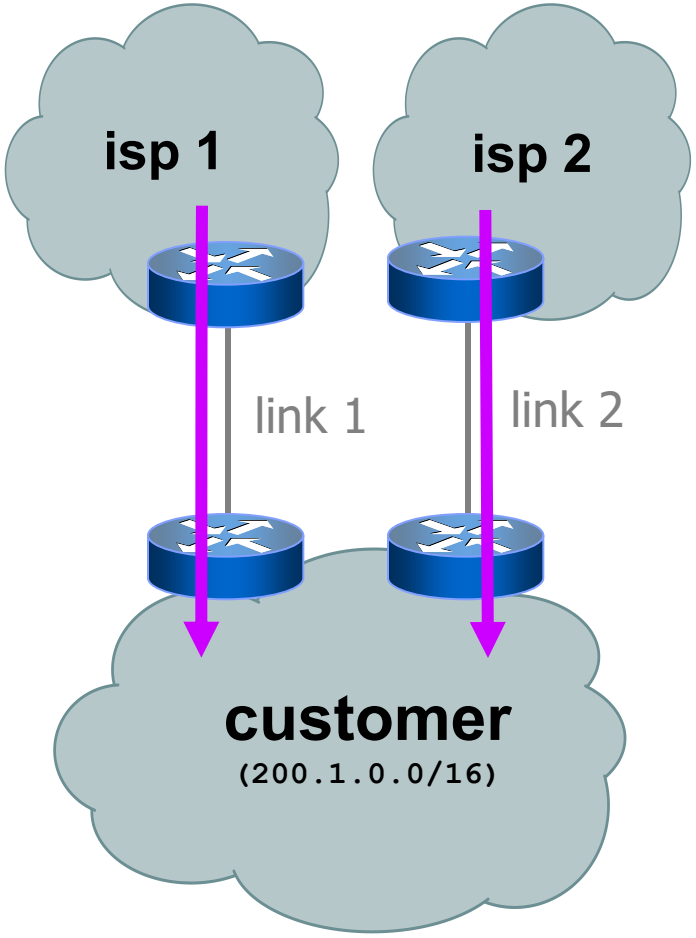
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet
- an inbound packet may use any of the two links in order to reach the network

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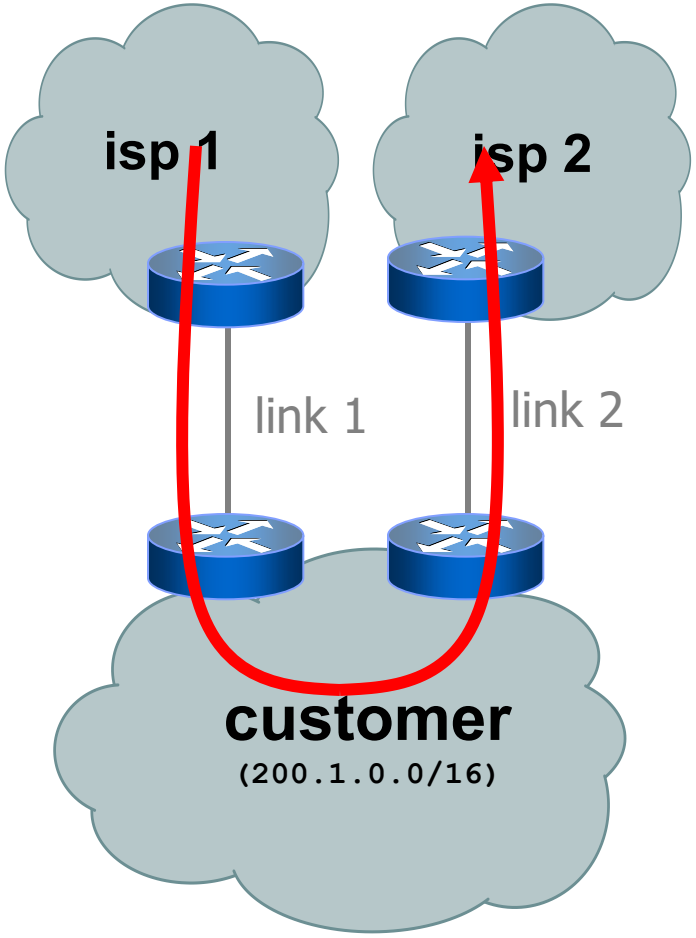
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet
- an inbound packet may use any of the two links in order to reach the network
- an internet packet may traverse link 1 and link 2 (or vice versa)

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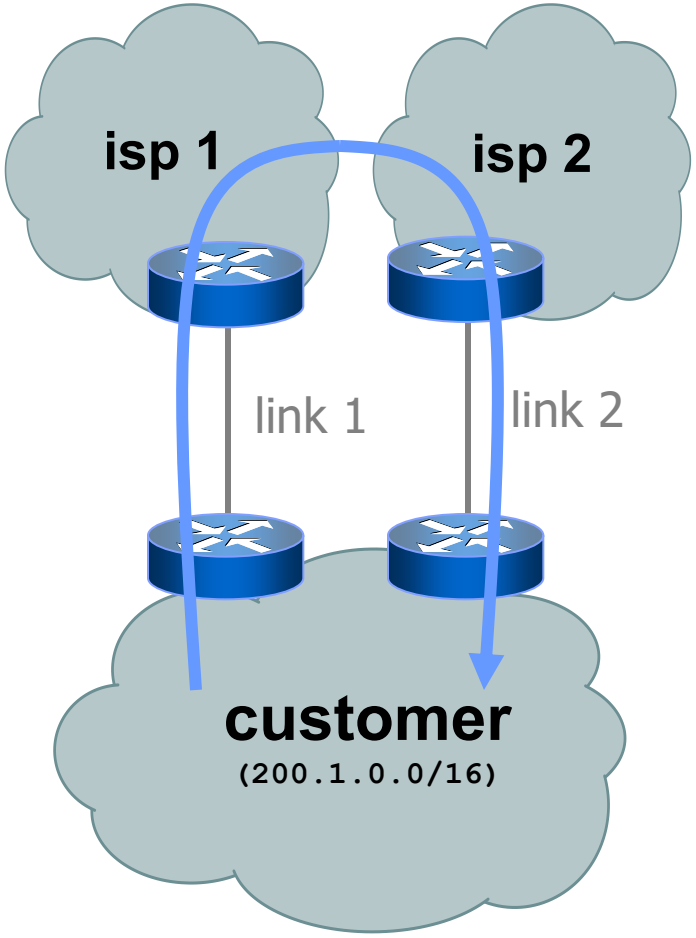
degrees of freedom



- an outbound packet may be sent through one of the two links in order to reach the internet
- an inbound packet may use any of the two links in order to reach the network
- an internet packet may traverse link 1 and link 2 (or vice versa)
- a local packet may traverse link 1 and link 2 (or vice versa)

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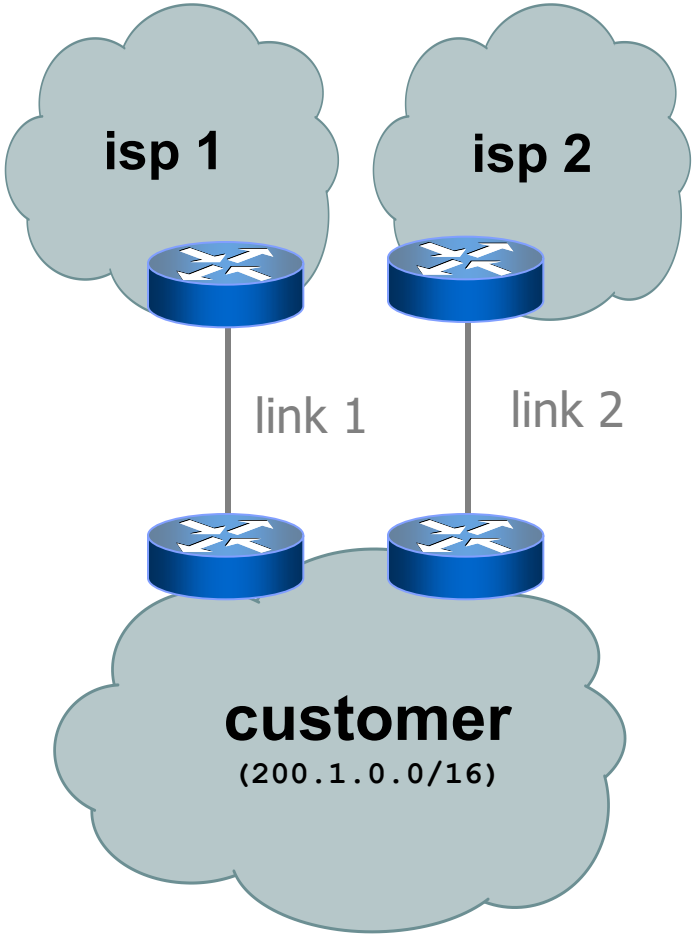
desired policy: loadsharing



- rule out transit flows
- outbound traffic:
  - half of the internal hosts use link 1
  - the other half uses link 2
- inbound traffic:
  - use link 1 when going to half the internal hosts
  - use link 2 when going to the other half

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using bgp for loadsharing

- announce /16 aggregate on each link
- split /16 and announce as two /17s, one on each link
  - rough loadsharing on inbound traffic
  - assumes equal circuit capacity and even spread of traffic across address block
- vary the split until "perfect" loadsharing achieved
- accept the default from upstream
  - basic outbound loadsharing by nearest exit ($\Rightarrow$ no local preference)
  - okay in first approximation as most customer traffic is inbound

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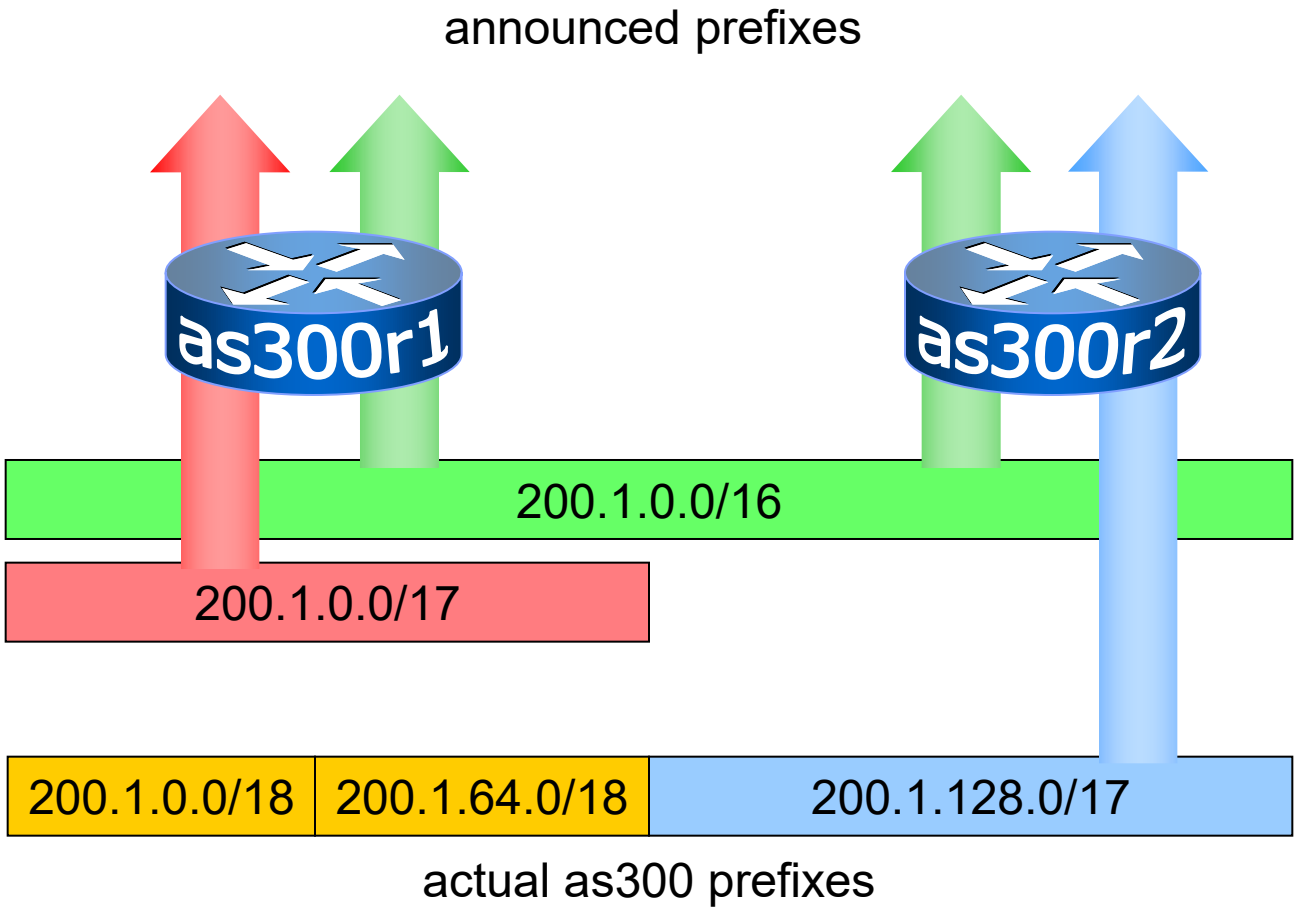
using bgp for loadsharing



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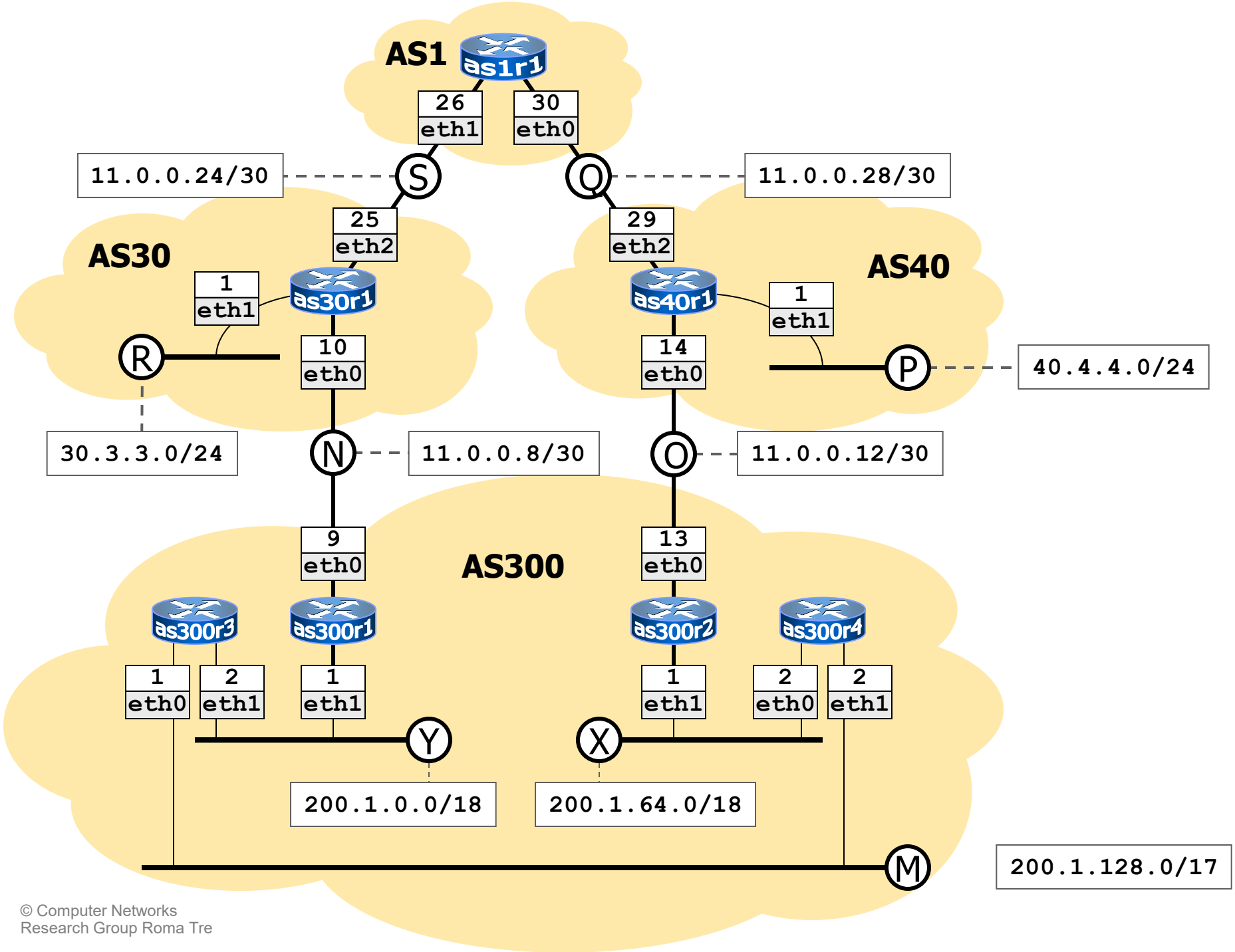

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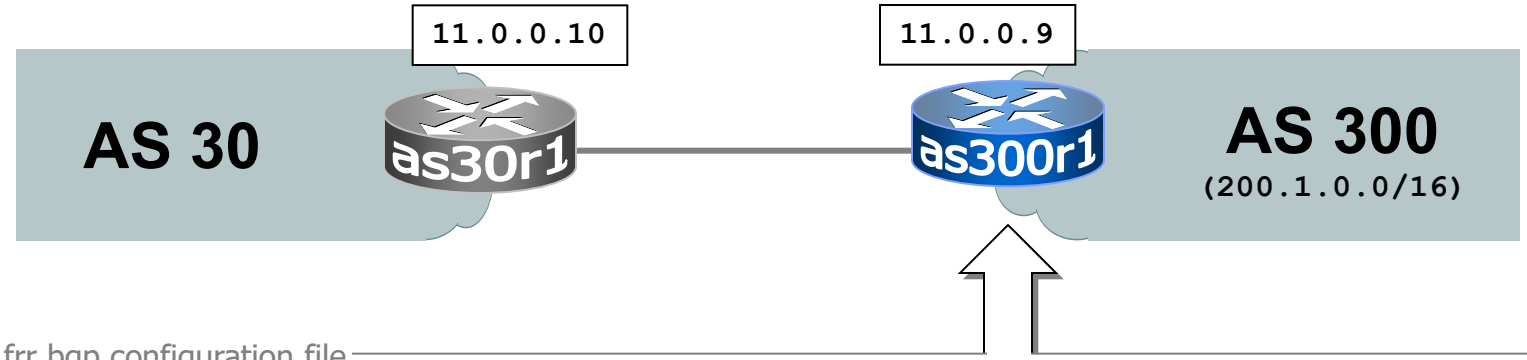
router as300r1 configuration



frr bgp configuration file

```
router bgp 300
no bgp network import-check
!
neighbor 11.0.0.10 remote-as 30
neighbor 11.0.0.10 description Router as30r1
!
network 200.1.0.0/16
network 200.1.0.0/17
!
... (continue)
```

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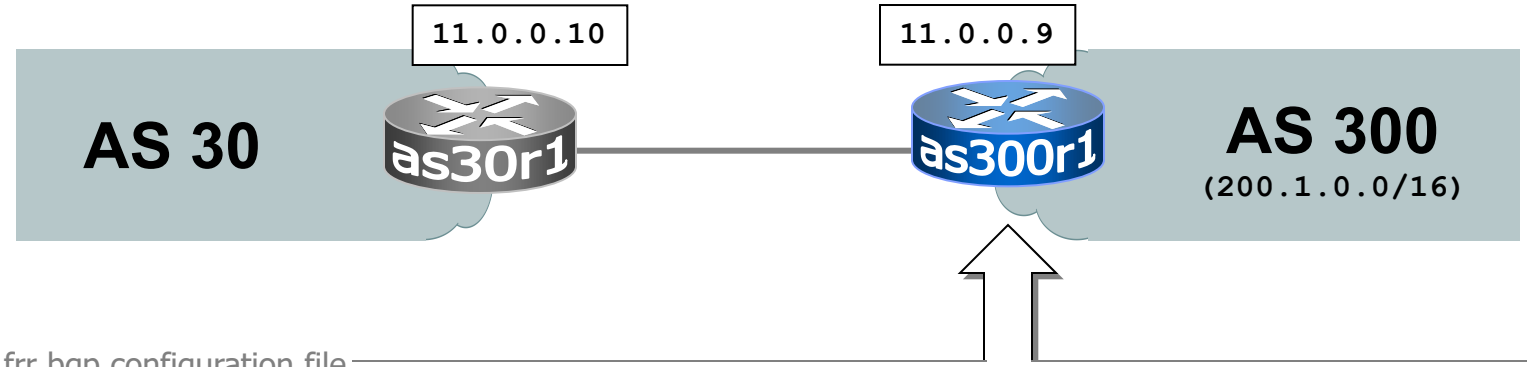
router as300r1 configuration



frr bgp configuration file

```
... (continue from previous page)
!
neighbor 11.0.0.10 prefix-list mineOutOnly out
neighbor 11.0.0.10 prefix-list defaultIn in
!
ip prefix-list mineOutOnly permit 200.1.0.0/16
ip prefix-list mineOutOnly permit 200.1.0.0/17
ip prefix-list defaultIn permit 0.0.0.0/0
```

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router as300r2 configuration

frr bgp configuration file

```
router bgp 300
no bgp network import-check
!
neighbor 11.0.0.14 remote-as 40
neighbor 11.0.0.14 description Router as40r1
!
network 200.1.0.0/16
network 200.1.128.0/17
!
... (continue)
```

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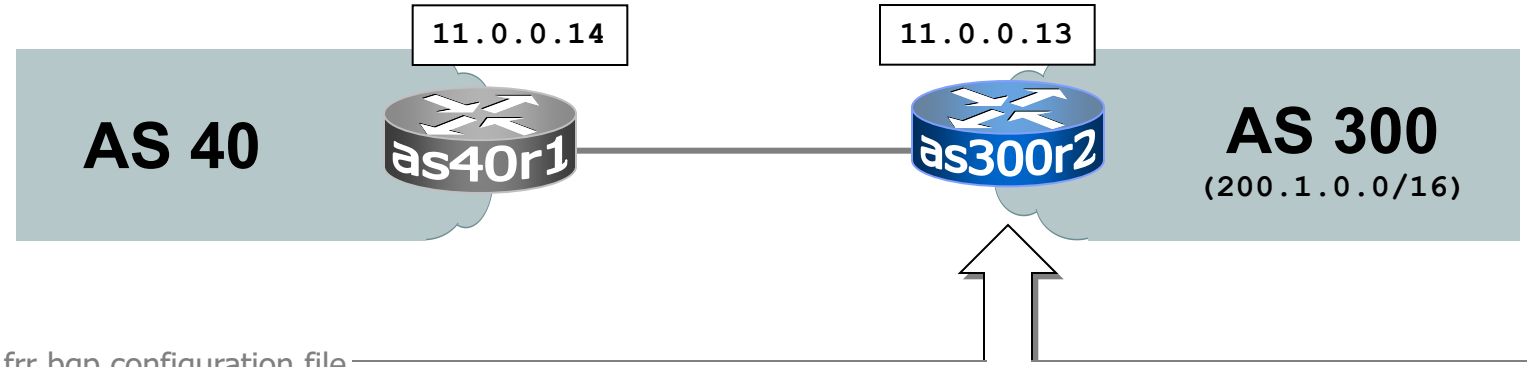
router as300r2 configuration



frr bgp configuration file

```
... (continue from previous page)
!
neighbor 11.0.0.14 prefix-list mineOutOnly out
neighbor 11.0.0.14 prefix-list defaultIn in
!
ip prefix-list mineOutOnly permit 200.1.0.0/16
ip prefix-list mineOutOnly permit 200.1.128.0/17
ip prefix-list defaultIn permit 0.0.0.0/0
```

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loadsharing

- experiment loadsharing

as1r1

```
root@as1r1:/# traceroute 200.1.0.2
traceroute to 200.1.0.2 (200.1.0.2), 64 hops max, 40 byte packets
 1  11.0.0.25 (11.0.0.25)  2 ms  2 ms  1 ms
 2  11.0.0.9 (11.0.0.9)  1 ms  2 ms  1 ms
 3  200.1.0.2 (200.1.0.2)  2 ms  3 ms  3 ms
root@as1r1:/# traceroute 200.1.128.2
traceroute to 200.1.128.2 (200.1.128.2), 64 hops max, 40 byte packets
 1  11.0.0.29 (11.0.0.29)  1 ms  2 ms  1 ms
 2  11.0.0.13 (11.0.0.13)  3 ms  2 ms  3 ms
 3  200.1.128.2 (200.1.128.2)  12 ms  3 ms  2 ms
root@as1r1:/#
```

- check the rip routing inside as300

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backup

- experiment backup
  - crash collision domain O as follows:

as300r2

```
root@as300r2:/# vtysh
.....
as300r2-frr# configure terminal
as300r2-frr(config)# router bgp 300
as300r2-frr(config-router)# neighbor 11.0.0.14 shutdown
as300r2-frr(config-router)# quit
as300r2-frr(config)# quit
as300r2-frr# quit
```

  - wait for rip to converge

as300r2

```
root@as300r2:/# route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         200.1.64.2      0.0.0.0         UG    20     0        0 eth1
11.0.0.12       0.0.0.0         255.255.255.252 U     0      0        0 eth0
200.1.0.0       200.1.64.2      255.255.192.0   UG    20     0        0 eth1
200.1.64.0      0.0.0.0         255.255.192.0   U     0      0        0 eth1
200.1.128.0     200.1.64.2      255.255.128.0   UG    20     0        0 eth1
```

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backup

- check the routing table of as1r1

as1r1

```
as1r1-frr# show ip bgp
...
...

   Network          Next Hop            Metric LocPrf Weight Path
*> 0.0.0.0/0        0.0.0.0                  0         32768 i
*> 11.0.0.8/30      11.0.0.25                0             0 30 i
*> 11.0.0.12/30     11.0.0.29                0             0 40 i
*> 11.0.0.24/30     0.0.0.0                  0         32768 i
*> 11.0.0.28/30     0.0.0.0                  0         32768 i
*> 30.3.3.0/24      11.0.0.25                0             0 30 i
*> 40.4.4.0/24      11.0.0.29                0             0 40 i
*> 200.1.0.0/16     11.0.0.25                              0 30 300 i
*> 200.1.0.0/17     11.0.0.25                              0 30 300 i

Displayed  9 routes and 9 total paths
as1r1-frr# quit
root@as1r1:/# traceroute 200.1.128.2
traceroute to 200.1.128.2 (200.1.128.2), 30 hops max, 60 byte packets
 1  11.0.0.25 (11.0.0.25)  0.260 ms  0.120 ms  0.119 ms
 2  11.0.0.9 (11.0.0.9)  0.236 ms  0.201 ms  0.225 ms
 3  200.1.0.2 (200.1.0.2)  0.301 ms  0.258 ms  0.258 ms
 4  200.1.128.2 (200.1.128.2)  0.392 ms
```

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